Topic: Reduction of order

Question: Use reduction of order to find the general solution to the differential equation, given $y_1 = e^x$.

$$y'' - 2y' + y = 0$$

Answer choices:

A $y(x) = c_1 e^x + c_2 e^{-x}$

B $y(x) = c_1 e^x + c_2 x e^x$

C $y(x) = c_1 e^x + c_2 e^{2x}$

D $y(x) = c_1 e^x + c_2 x$

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Solution: B

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = ve^x$$

$$y_2' = v'e^x + ve^x$$

$$y_2'' = v''e^x + 2v'e^x + ve^x$$

Plug these values into the homogeneous differential equation.

$$v''e^x + 2v'e^x + ve^x - 2(v'e^x + ve^x) + ve^x = 0$$

$$v''e^x = 0$$

We'll make a substitution $w = v'$, and therefore $w' = v''$.

$$w'e^x = 0$$

This substitution changes the second order equation in v into a first order linear equation in w that we can easily solve.

$$e^x \frac{dw}{dx} = 0$$

$$\frac{dw}{dx} = 0$$

$$w = C$$

Because $w = v'$, we'll integrate w to find v.

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$$v = \int w \, dx = \int C \, dx = Cx + k$$

We can choose any constants, so we'll choose $C = 1$ and $k = 0$ for simplicity. Then v is

$$v = x$$

and the second solution for the differential equation is

$$y_2 = ve^x$$

$$y_2 = xe^x$$

Because the solutions are $y_1 = e^x$ and $y_2 = xe^x$, the general solution to the differential equation is

$$y(x) = c_1 e^x + c_2 xe^x$$

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Topic: Reduction of order

Question: Use reduction of order to find the general solution to the differential equation, given $y_1 = \cos x^2$.

$$y'' - \frac{1}{x}y' + 4x^2y = 0$$

Answer choices:

A $y = c_1 \cos x^2 + c_2 x \sin x^2$

B $y = c_1 \cos x^2 + c_2 \sin x$

C $y = e^x(c_1 \cos x^2 + c_2 \sin x^2)$

D $y = c_1 \cos x^2 + c_2 \sin x^2$

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Solution: D

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = v \cos x^2$$

$$y_2' = v' \cos x^2 - 2xv \sin x^2$$

$$y_2'' = v'' \cos x^2 - 4xv' \sin x^2 - 2v \sin x^2 - 4vx^2 \cos x^2$$

Plug these values into the homogeneous differential equation.

$$v'' \cos x^2 - 4xv' \sin x^2 - 2v \sin x^2 - 4vx^2 \cos x^2$$

$$-\frac{1}{x}(v' \cos x^2 - 2xv \sin x^2) + 4x^2 v \cos x^2 = 0$$

$$v'' \cos x^2 - 4xv' \sin x^2 - 2v \sin x^2 - 4vx^2 \cos x^2$$

$$-v'\frac{1}{x} \cos x^2 + 2v \sin x^2 + 4x^2 v \cos x^2 = 0$$

$$v'' \cos x^2 - v' \left(4x \sin x^2 + \frac{1}{x} \cos x^2 \right) = 0$$

We'll make a substitution $w = v'$, and therefore $w' = v''$.

$$w' \cos x^2 - w \left(4x \sin x^2 + \frac{1}{x} \cos x^2 \right) = 0$$

This substitution changes the second order equation in v into a first order linear equation in w that we can easily solve.

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$$\cos x^2 \frac{dw}{dx} - w\left(4x \sin x^2 + \frac{1}{x}\cos x^2\right) = 0$$

$$\frac{dw}{dx} - w\left(\frac{4x \sin x^2}{\cos x^2} + \frac{\cos x^2}{x \cos x^2}\right) = 0$$

$$\frac{dw}{dx} - w\left(4x \tan x^2 + \frac{1}{x}\right) = 0$$

$$\frac{1}{w}\, dw = \left(4x \tan x^2 + \frac{1}{x}\right)\, dx$$

$$\int \frac{1}{w}\, dw = \int 4x \tan x^2 + \frac{1}{x}\, dx$$

$$\ln w = \ln x + C + \int 4x \tan x^2\, dx$$

Let $t = x^2$ and $dt = 2x\ dx$.

$$\ln w = \ln x + C + 2\int \tan t\ dt$$

$$\ln w = \ln x + C - 2 \ln \cos t$$

$$\ln w = \ln x - 2 \ln \cos x^2 + C$$

$$\ln w = \ln \frac{x}{\cos^2 x^2} + C$$

$$w = C\frac{x}{\cos^2 x^2}$$

Because $w = v'$, we'll integrate w to find v.

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$$v = \int w\ dx = \int C\frac{x}{\cos^2 x^2}\ dx$$

Let $t = x^2$ and $dt = 2x\ dx$.

$$v = \int w\ dx = \frac{C}{2}\int \frac{1}{\cos^2 t}\ dt$$

$$v = \frac{C}{2}\tan t + k$$

$$v = \frac{C}{2}\tan x^2 + k$$

We can choose any constants, so we'll choose $C = 2$ and $k = 0$ for simplicity. Then v is

$$v = \tan x^2$$

and the second solution for the differential equation is

$$y_2 = v\cos x^2$$

$$y_2 = \tan x^2 \cos x^2$$

$$y_2 = \sin x^2$$

Because the solutions are $y_2 = \cos x^2$ and $y_2 = \sin x^2$, the general solution to the differential equation is

$$y(x) = c_1 \cos x^2 + c_2 \sin x^2$$

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Topic: Reduction of order

Question: Use reduction of order to find the general solution to the differential equation, given $y_1 = \sin x$.

$$y'' + y = 0$$

Answer choices:

A $y = e^x(c_1 \cos x + c_2 \sin x)$

B $y = e^{-x}(c_1 \cos x + c_2 \sin x)$

C $y = c_1 \sin x + c_2 \cos x$

D $y = c_1 \sin x + c_2 x \cos x$

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Solution: C

To apply this method, we always begin with the assumption that $y_2 = vy_1$, and then we find the first and second derivatives of y_2, since we're dealing with a second order equation.

$$y_2 = v \sin x$$

$$y_2' = v' \sin x + v \cos x$$

$$y_2'' = v'' \sin x + 2v' \cos x - v \sin x$$

Plug these values into the homogeneous differential equation.

$$v'' \sin x + 2v' \cos x - v \sin x + v \sin x = 0$$

$$v'' \sin x + 2v' \cos x = 0$$

We'll make a substitution $w = v'$, and therefore $w' = v''$.

$$w' \sin x + 2w \cos x = 0$$

This substitution changes the second order equation in v into a first order linear equation in w that we can easily solve.

$$\sin x \frac{dw}{dx} + 2w \cos x = 0$$

$$\frac{dw}{dx} + \frac{2 \cos x}{\sin x} w = 0$$

$$\frac{dw}{dx} + (2 \cot x) w = 0$$

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With $P(x) = 2\cot x$ and $Q(x) = 0$, the integrating factor is

$$I(x) = e^{\int P(x)\ dx}$$

$$I(x) = e^{\int 2\cot x\ dx}$$

$$I(x) = e^{2\ln\sin x}$$

$$I(x) = \sin^2 x$$

Multiplying the first order differential equation in w by the integrating factor, and then simplifying, gives

$$\frac{d}{dx}(w\sin^2 x) = 0$$

Integrate both sides.

$$\int \frac{d}{dx}(w\sin^2 x)\ dx = \int 0\ dx$$

$$w\sin^2 x = C$$

$$w = \frac{C}{\sin^2 x}$$

Because $w = v'$, we'll integrate w to find v.

$$v = \int w\ dx = \int \frac{C}{\sin^2 x}\ dx = -C\cot x + k$$

We can choose any constants, so we'll choose $C = -1$ and $k = 0$ for simplicity. Then v is

$$v = \cot x$$

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and the second solution for the differential equation is

$$y_2 = \cot x \sin x$$

$$y_2 = \cos x$$

Because the solutions are $y_1 = \sin x$ and $y_2 = \cos x$, the general solution to the differential equation is

$$y(x) = c_1 \sin x + c_2 \cos x$$

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